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ASSIGNMENT-



Subject: Database Management System

Subject Code: 24CSH-298

Branch Sec: KRG1-G2CA7

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Assignment-2

Q1) Explain each type of database security issues.

Ans) Database security issues are generally of 4 main categories.

1) Injection Attacks: Malicious code is used to trick the database into revealing private data or deleting records.

2) Human Factor: This includes insider threats, weak passwords or accidentally leaving sensitive information exposed.

3) Broken Access Control: Giving users excessive privileges they don't need. It creates a massive backdoor if their account is hacked.

4) System Weakness: Using unpatched software, default "admin" passwords or failing to encrypt backups and data in transit.

Q2) Explain different database security control techniques.

Ans) Database security controls are broken into three simple categories.

1) Technical Software

- Access Control: Verifies (MFA), Limits action (RBAC)
- Encryption: Scrambles data.
- Masking: Hide sensitive details
- Firewalls: Blocks malicious queries.

2) Administrative

- Least Privilege: Users get minimum access required.
- Auditing: Keeping record of every login
- Patching: Regularly updating software.
- Backups: Keeping encrypted copies to recover from ransomware.

3) Physical

- Site security: Using locks, cameras, and guards to protect servers.
- Redundancy: Protecting hardware from fire, floods, or power outages.

Q3) Consider a table Employee (EmpId, EmpName, Salary, DeptId). Write a SQL query to create a view that displays only EmpId, EmpName and DeptId from the Employee table.

Ans) This SQL query creates a view, which acts as a virtual table to filter what data a user can see.

SQL Query:

```
CREATE VIEW Employee-view AS  
SELECT EmpId, EmpName, DeptID  
FROM Employee;
```